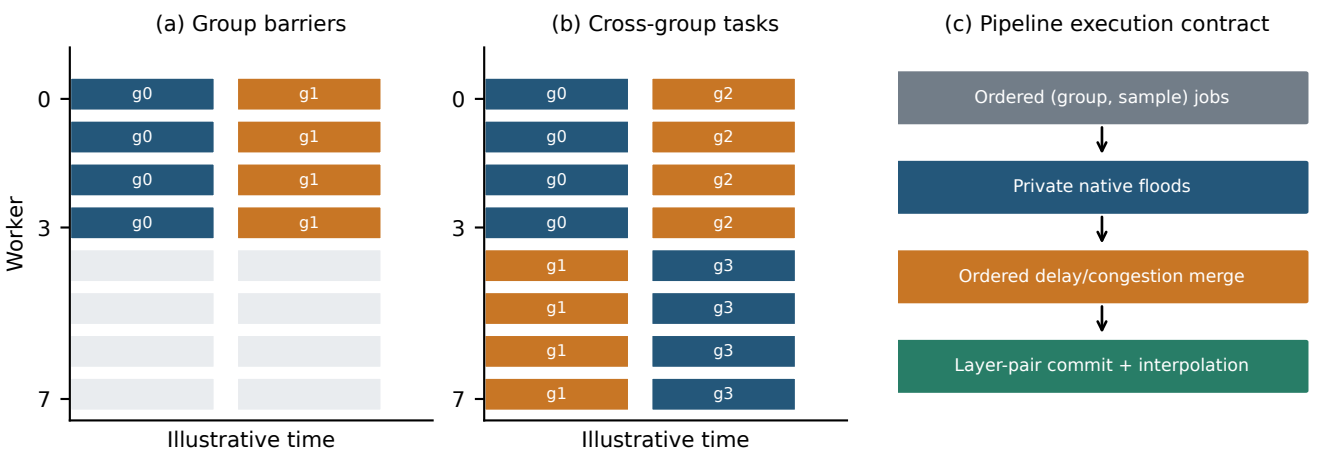
(a) Group barriers
Worker
0
3
7
g0
g1
Illustrative time
(b) Cross-group tasks
0
3
7
g0
g2
g1
g3
Illustrative time
(c) Pipeline execution contract
Ordered (group, sample) jobs
Private native floods
Ordered delay/congestion merge
Layer-pair commit + interpolation